Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 10.1 · Convert and Measure

The unit circle is used to measure angles in radians.

1. Convert 150 $^\circ$ & -90° into radians, showing the conversion factor.
2. A circular sector has a radius of 4 cm and an angle of $\frac{\pi}{3}$ radians. Calculate the exact length of its arc.
3. Explain why an angle measure can be negative while a geometric arc length is positive.

Exercise 10.2 · Reduce angles

Consider the real numbers $\frac{17\pi}{6}$, $-\frac{7\pi}{4}$ and 5π.

1. For each, determine a representative in [0; 2 π[having the same image on the unit circle. Show the number of turns added or removed.
2. Plot the three images on a circle and label them with the original real numbers.

Exercise 10.3 · Exact values

We look for the exact values of the cosine and sine of $\frac{2\pi}{3}$, $-\frac{\pi}{6}$ and $\frac{5\pi}{4}$.

1. Place the three images on the unit circle. For each, state the expected sign of its two coordinates.
2. Calculate the exact values using a standard angle and a symmetry. Present the answer in a table: angle, cosine, sine.

Exercise 10.4 · Find the sign

A real x satisfies $\sin x = \frac{5}{13}$. Its image on the unit circle lies in the second quadrant, hence to the left of the y-axis and above the x-axis.

1. Determine the exact value of cos x. Justify both its absolute value and its sign.

2. Explain why knowing only sin x would not have been enough to choose the sign of cos x.

Exercise 10.5 · A wheel

A wheel of radius 0 ,30 m rolls on horizontal ground without slipping. It turns through an angle $\frac{5\pi}{3}$ radians. We want the distance travelled by its axle in the direction of motion.

1. Calculate this distance exactly, explaining the link with the unrolled arc length.

2. Explain why the length of a chord of the circle must not be used. Is the path of a fixed point on the rim of the wheel, seen from the ground, the same as that of the axle?

Exercise 10.6 · Chord and scalar product

On the unit circle with centre O, the point A is the image of 0 and the point B that of $\frac{2\pi}{3}$.

1. Determine the exact coordinates of A and B, then calculate the length AB.

2. Find this length again by applying Al-Kashi's formula to the triangle OAB. State the lengths and the angle used.

Exercise 10.7 · Read an equation on the circle

We look for the real numbers x in the interval [0; 2 π[whose image on the unit circle has x-coordinate $\frac{1}{2}$.

1. Express the condition on the x-coordinate as an equation involving cos x.

2. Draw the corresponding vertical line on a circle, locate its points of intersection and give the two solutions. Check that they belong to the required interval.

Exercise 10.8 · Delayed review

Consider the angle with measure $\frac{25\pi}{4}$ radians. Answers must be exact, without decimal approx.

1. Reduce this angle by removing a whole number of complete turns.

2. Deduce its cosine and sine. Justify the values using the unit circle.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 32 p. 220, questions a and b. Use the exact value given and the associated angles, without a calculator.

CQFD, Exercise 39 p. 220, Questions 1, 2, 3 and 5. Do the four questions on calculation and reading the circle; justify the answers on your script.